1. SECURITY AUDIT OF A COMPUTER SYSTEM

Windows Host Security Posture & Compliance Report: LAB-DEMO-PC01 (DEMO SAMPLE DATA)

Comprehensive Read-Only Assessment of User Accounts, System Updates, File Permissions, Running Services, and Security Configurations.

2–4. Target Computer Information, Operating System Details & Audit Metadata

2. Computer Information	Value	3. OS & 4. Audit Details	Value
Hostname	LAB-DEMO-PC01 (DEMO SAMPLE DATA)	**Operating System**	Microsoft Windows 11 Home
Manufacturer / Model	University Cyber Lab Demonstration Workstation	**OS Version & Build**	10.0.26200 (Build 26200)
Processor (CPU)	AMD Ryzen 7 7735HS with Radeon Graphics (16 Logical Cores)	**OS Architecture**	64-bit
Installed Memory (RAM)	15.19 GB	**PowerShell Version**	5.1.26100.9711
Domain / Workgroup	Workgroup: WORKGROUP	**Audit Session ID**	DEMO-SAMPLE-2026
Audited User Context	demo_student (Standard User)	**Audit Date & Time (UTC)**	2026-11-05 09:24:33 (26.36s)

5. Executive Summary

The Windows Security Audit evaluated 31 security controls across 5 core domains, achieving an overall security score of 86.0/100 (Elevated Risk (Action Required)). In total, 25 checks passed, 3 failed, 2 raised warnings, and 1 could not be assessed under the current privilege level. Key areas requiring attention include 1 Critical and 2 High severity findings.

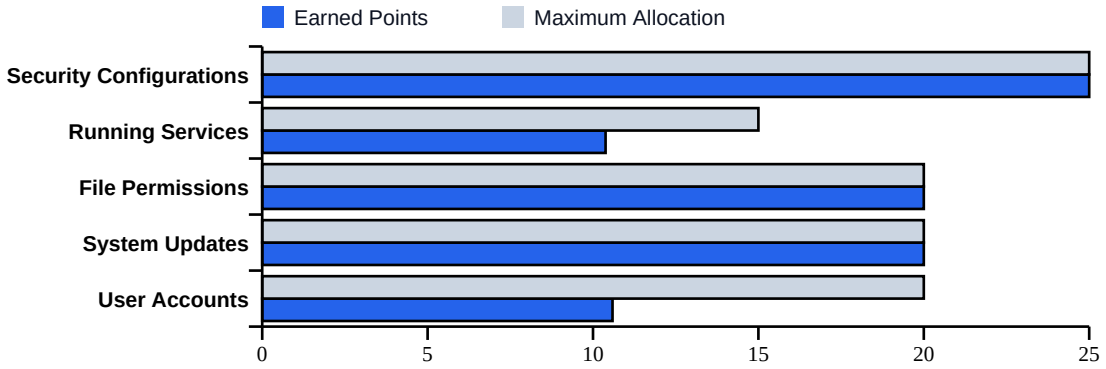
6. Overall Security Score & Category Breakdown

OVERALL SECURITY SCORE: 86.0 / 100

Risk Posture Rating: **Elevated Risk (Action Required)**

Total Checks: 31
Passed: 25 | **Failed:** 3
Warnings: 2 | **Not Assessed:** 1
Critical Findings: 1 | **High Findings:** 2

Audit Category	Earned / Max	Score %	Passed	Failed	Warnings	Not Assessed
User Accounts	10.59 / 20.0	52.9%	5	2	0	0
System Updates	20.0 / 20.0	100.0%	5	0	0	0
File Permissions	20.0 / 20.0	100.0%	5	0	0	0
Running Services	10.38 / 15.0	69.2%	2	1	2	0
Security Configurations	25.0 / 25.0	100.0%	8	0	0	1



Methodology: Scoring Allocation: User Accounts (20 pts), System Updates (20 pts), File Permissions (20 pts), Running Services (15 pts), Security Configurations (25 pts). Severity Weights: Critical=10, High=6, Medium=3, Low=1. PASS awards 100% of check weight, WARNING awards 50%, FAIL awards 0%. Checks marked NOT_ASSESSED (e.g., due to non-administrative privilege context) are excluded from the denominator.

7. User Account Audit Findings (Module B)

Inspected **5** local user accounts (1 Enabled, 1 Administrator). Minimum Password Length: **0 chars** | Lockout Threshold: **10**.

ID & Check	Status / Severity	Evidence & Explanation	Recommended Remediation
USR-001 Built-in Guest Account Status [Configuration Observation]	**PASS** Severity: **High**	**Evidence:** `Account 'Guest' (SID: S-1-5-21-1301171905-3339115337-1909463598-501) is Disabled (Enabled = False).` **Impact:** Disabling the Guest account prevents anonymous local and network access.	No action required. Maintain Guest account in disabled state.
USR-002 Built-in Administrator Account (RID 500) Status [Configuration Observation]	**PASS** Severity: **Medium**	**Evidence:** `Built-in Administrator 'Administrator' (SID: S-1-5-21-1301171905-3339115337-1909463598-500) is Disabled.` **Impact:** Keeping the built-in Administrator account disabled reduces the attack surface for RID-500 brute-force attacks.	No action required.
USR-003 Administrative Privilege Minimization [Configuration Observation]	**PASS** Severity: **Medium**	**Evidence:** `1 enabled Administrator account(s) detected: demo_student.` **Impact:** Restricting administrative privileges to a single administrative identity adheres to the Principle of Least Privilege.	Continue using Standard User accounts for routine non-administrative tasks.
USR-004 Password Requirement on Enabled Accounts [Confirmed Vulnerability]	**FAIL** Severity: **Critical**	**Evidence:** `Enabled account(s) with PasswordRequired=False: demo_student` **Impact:** Accounts that do not require a password allow trivial unauthorized local login and privilege escalation.	Set a strong password and enforce PasswordRequired on all enabled local user accounts.
USR-005 Password Expiration Policy on Active Accounts [Configuration Observation]	**PASS** Severity: **Medium**	**Evidence:** `All enabled user accounts enforce password expiration.` **Impact:** Periodic credential rotation limits the window of exposure for leaked credentials.	No action required.
USR-006 System Minimum Password Length Policy [Security Weakness]	**FAIL** Severity: **High**	**Evidence:** `net accounts reports Minimum password length: 0 characters.` **Impact:** A minimum password length of 0 permits blank or short passwords that can be cracked instantaneously.	Configure Minimum Password Length to at least 12 characters via secpol.msc or 'net accounts /minpwlen:12' (as Administrator).
USR-007 Account Lockout Threshold Policy [Configuration Observation]	**PASS** Severity: **Medium**	**Evidence:** `net accounts reports Lockout threshold: 10 invalid attempts (Duration: 10 mins).` **Impact:** An active account lockout policy automatically throttles automated password-spraying and brute-force attacks.	No action required.

8. System Update Audit Findings (Module C)

OS Build: **Microsoft Windows 11 Home (Build 26200)** | Installed Hotfixes: **5** | Latest Update Date: **2026-10-28** (8 days ago).

ID & Check	Status / Severity	Evidence & Explanation	Recommended Remediation
UPD-001 Operating System Build Support & Currency [Configuration Observation]	**PASS** Severity: **High**	**Evidence:** `Microsoft Windows 11 Home (Version 10.0.26200, Build 26200) is a modern supported Windows 11 servicing build.` **Impact:** Running an actively supported Windows build ensures the kernel and system components receive monthly security patches.	No action required. Maintain current feature update servicing channel.

ID & Check	Status / Severity	Evidence & Explanation	Recommended Remediation
UPD-002 Installed Security Updates & Patch Recency [Configuration Observation]	**PASS** Severity: **High**	**Evidence:** `5 hotfixes installed. Latest update KB5127648 installed on 2026-10-28 (8 days ago).` **Impact:** Microsoft releases cumulative security updates monthly (Patch Tuesday). Updates within the last 45 days indicate healthy patch hygiene.	Continue applying monthly cumulative updates.
UPD-003 Windows Update Services Availability [Configuration Observation]	**PASS** Severity: **Medium**	**Evidence:** `wuauserv: Running (Manual), UsoSvc: Running (Automatic)` **Impact:** Windows Update services are enabled (Manual Trigger-Start or Automatic is standard on Windows 10/11).	No action required.
UPD-004 Automatic Updates Configuration [Configuration Observation]	**PASS** Severity: **High**	**Evidence:** `Automatic Updates policy is active (NoAutoUpdate=0).` **Impact:** Automatic updates ensure critical OS and Defender security patches are delivered promptly.	No action required.
UPD-005 Pending Windows Updates Queue [Configuration Observation]	**PASS** Severity: **Medium**	**Evidence:** `0 pending software updates found in the local Windows Update cache.` **Impact:** No uninstalled updates are waiting in the local update queue.	No action required.

9. File and Directory Permission Findings (Module D)

Inspected NTFS Access Control Lists across **6** critical system directories (48 ACEs evaluated; **0** risky ACEs flagged).

ID & Check	Status / Severity	Evidence & Explanation	Recommended Remediation
PERM-001 Overly Broad 'Everyone' Write/FullControl Permissions [Configuration Observation]	**PASS** Severity: **Critical**	**Evidence:** `0 audited directories or files grant Write, Modify, or FullControl to 'Everyone'.` **Impact:** Preventing 'Everyone' write access protects system integrity against unauthorized tampering.	No action required.
PERM-002 System32 and Program Files Write Protection [Configuration Observation]	**PASS** Severity: **High**	**Evidence:** `C:\Windows\System32 and C:\Program Files restrict write/modify access to TrustedInstaller, SYSTEM, and Administrators.` **Impact:** Strict ACLs on OS and application binary directories prevent local privilege escalation via binary replacement.	No action required.
PERM-003 All-Users Startup Directory Permission Hardening [Configuration Observation]	**PASS** Severity: **High**	**Evidence:** `Machine-wide Startup folder prevents write/modify operations by standard users.` **Impact:** Securing the global Startup folder prevents cross-account persistence and privilege escalation.	No action required.
PERM-004 Windows DNS Hosts File ACL Protection [Configuration Observation]	**PASS** Severity: **High**	**Evidence:** `C:\Windows\System32\drivers\etc\hosts is read-only for standard users and writable only by Administrators/SYSTEM.` **Impact:** Protecting the hosts file prevents local DNS poisoning and security update redirection.	No action required.
PERM-005 Shared & Temporary Directory ACL Review [Configuration Observation]	**PASS** Severity: **Medium**	**Evidence:** `C:\Users\Public and C:\Windows\Temp follow standard scoped NTFS inheritance without broad FullControl exposure.` **Impact:** Properly scoped permissions on shared directories prevent cross-user file tampering.	No action required.

10. Running Services Audit Findings (Module E)

Total Services: **289** | Running: **135** | Running as LocalSystem: **91** | Flagged Risky Services: **2**.

ID & Check	Status / Severity	Evidence & Explanation	Recommended Remediation
SVC-001 Unquoted Service Binary Paths (CWE-428) [Confirmed Vulnerability]	**FAIL** Severity: **High**	**Evidence:** `Detected 1 service(s) with unquoted paths containing spaces: UpcElevationService (C:\Program Files (x86)\Ubisoft\Ubisoft Game Launcher Core\UpcElevationService.exe)` **Impact:** When a service path contains spaces and is unquoted, Windows attempts to execute binaries at each space boundary (e.g. C:\Program.exe), enabling privilege escalation if a parent folder is writable.	Wrap the ImagePath registry value in double quotes for each affected service in HKLM:\SYSTEM\CurrentControlSet\Services.
SVC-002 Remote Registry Service Hardening [Configuration Observation]	**PASS** Severity: **High**	**Evidence:** `RemoteRegistry is Stopped (StartMode: Disabled).` **Impact:** Keeping Remote Registry stopped prevents remote registry enumeration and lateral movement.	No action required.
SVC-003 Insecure Legacy Network Services (Telnet, FTP, SNMP, Simple TCP) [Configuration Observation]	**PASS** Severity: **Critical**	**Evidence:** `Telnet (TlntSvr), FTP (FTPSVC), SNMP, and Simple TCP/IP services are not running.` **Impact:** Disabling legacy cleartext daemons eliminates network credential sniffing and service exploitation vectors.	No action required.
SVC-004 Print Spooler Service Status [Configuration Observation]	**WARNING** Severity: **Low**	**Evidence:** `Print Spooler (Spooler) is Running as 'LocalSystem' (StartMode: Auto).` **Impact:** Print Spooler runs as LocalSystem by default. While needed for local printing, it has historically been a frequent vector for privilege escalation (e.g., PrintNightmare).	If printing is not required on this machine, disable the Spooler service or disable 'Allow Print Spooler to accept client connections' in Group Policy.
SVC-005 Third-Party Services Running as LocalSystem [Security Weakness]	**WARNING** Severity: **Medium**	**Evidence:** `15 third-party services are running as LocalSystem (including: LenovoVantageService, DAService, EABackgroundService, GameInputRedistService, GamingServices, HidHideWatchdog.exe). Total LocalSystem running services: 91/135.` **Impact:** Every third-party service running as LocalSystem represents full OS compromise if a vulnerability exists in that vendor daemon.	Audit third-party background services and uninstall unused software or downgrade service accounts to LocalService where supported.

11. Security Configuration Audit Findings (Module F)

Evaluated **9** core Windows security controls across Defender, Firewall, UAC, Remote Desktop, Secure Boot, BitLocker, and SMBv1.

ID & Check	Status / Severity	Evidence & Explanation	Recommended Remediation
SEC-001 Windows Defender Antivirus & Real-Time Protection [Configuration Observation]	**PASS** Severity: **Critical**	**Evidence:** `AntivirusEnabled=True, RealTimeProtectionEnabled=True, BehaviorMonitorEnabled=True, IoavProtectionEnabled=True` **Impact:** Real-Time Protection scans files and process memory on access to block malware execution before infection occurs.	Enable Real-time protection in Windows Security > Virus & threat protection > Manage settings.

ID & Check	Status / Severity	Evidence & Explanation	Recommended Remediation
SEC-002 Windows Defender Security Intelligence Currency [Configuration Observation]	**PASS** Severity: **High**	**Evidence:** `Signature Version: 1.463.134.0, AntivirusSignatureAge: 0 day(s).` **Impact:** Outdated antivirus signatures reduce detection accuracy against newly emerging ransomware and trojans.	Run 'Update-MpSignature' in PowerShell or check for protection updates in Windows Security.
SEC-003 Windows Defender Tamper Protection [Configuration Observation]	**PASS** Severity: **Medium**	**Evidence:** `IsTamperProtected = True` **Impact:** Tamper Protection prevents malware or scripts from disabling Windows Defender via registry or PowerShell.	Enable Tamper Protection in Windows Security > Virus & threat protection > Manage settings.
SEC-004 Windows Defender Firewall Profile Status (Domain, Private, Public) [Configuration Observation]	**PASS** Severity: **Critical**	**Evidence:** `Domain: Enabled=True (In:NotConfigured), Private: Enabled=True (In:NotConfigured), Public: Enabled=True (In:NotConfigured)` **Impact:** The host firewall blocks unsolicited inbound network connections and lateral movement attempts.	Enable all firewall profiles via 'Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled True' (as Administrator).
SEC-005 User Account Control (UAC) & Secure Desktop Configuration [Configuration Observation]	**PASS** Severity: **High**	**Evidence:** `EnableLUA=1, ConsentPromptBehaviorAdmin=5, PromptOnSecureDesktop=1.` **Impact:** UAC prevents silent privilege escalation by requiring interactive user consent on an isolated Secure Desktop.	Set UAC to 'Notify me only when apps try to make changes to my computer (default)' or higher in User Account Control Settings.
SEC-006 Remote Desktop Protocol (RDP) & Network Level Authentication [Configuration Observation]	**PASS** Severity: **High**	**Evidence:** `fDenyTSConnections = 1 (Remote Desktop connections are disabled).` **Impact:** Exposed RDP without Network Level Authentication is one of the primary initial access vectors for ransomware operators.	Disable Remote Desktop if not needed, or enforce Network Level Authentication (UserAuthentication = 1) and restrict TCP 3389 via firewall.
SEC-007 UEFI Secure Boot Integrity Protection [Configuration Observation]	**PASS** Severity: **High**	**Evidence:** `HKLM:\SYSTEM\CurrentControlSet\Control\SecureBoot\State\UEFISecureBootEnabled = 1` **Impact:** UEFI Secure Boot cryptographically verifies the OS bootloader before execution, blocking bootkits.	Enable Secure Boot in the system UEFI/BIOS firmware settings.
SEC-008 BitLocker Full Disk Encryption (OS Volume C:) [Configuration Observation]	**NOT_ASSESSED** Severity: **High**	**Evidence:** `Access denied` **Impact:** Full-disk encryption protects data at rest if a laptop or workstation is physically lost or stolen.	Run WinSecAudit as Administrator to assess BitLocker, or enable BitLocker in Control Panel > BitLocker Drive Encryption.
SEC-009 SMBv1 Legacy Protocol Deprecation [Configuration Observation]	**PASS** Severity: **Critical**	**Evidence:** `LanmanServer SMB1=0, mrxsmb10 driver state=NotInstalled.` **Impact:** SMBv1 lacks modern cryptographic integrity protections and was exploited by WannaCry and NotPetya worms.	Disable SMBv1 via 'Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol'.

12 & 13. Identified Weaknesses & Prioritized Remediation Roadmap

Priority / Check ID	Classification	Identified Weakness / Vulnerability	Actionable Remediation Step
Critical USR-004 (FAIL)	Confirmed Vulnerability	**Password Requirement on Enabled Accounts:** Enabled account(s) with PasswordRequired=False: demo_student	Set a strong password and enforce PasswordRequired on all enabled local user accounts.
High USR-006 (FAIL)	Security Weakness	**System Minimum Password Length Policy:** net accounts reports Minimum password length: 0 characters.	Configure Minimum Password Length to at least 12 characters via secpol.msc or 'net accounts /minpwlen:12' (as Administrator).

Priority / Check ID	Classification	Identified Weakness / Vulnerability	Actionable Remediation Step
High SVC-001 (FAIL)	Confirmed Vulnerability	**Unquoted Service Binary Paths (CWE-428):** Detected 1 service(s) with unquoted paths containing spaces: UpcElevationService (C:\Program Files (x86)\Ubisoft\Ubisoft Game Launcher Core\UpcElevationService.exe)	Wrap the ImagePath registry value in double quotes for each affected service in HKLM:\SYSTEM\CurrentControlSet\Services.
Low SVC-004 (WARNING)	Configuration Observation	**Print Spooler Service Status:** Print Spooler (Spooler) is Running as 'LocalSystem' (StartMode: Auto).	If printing is not required on this machine, disable the Spooler service or disable 'Allow Print Spooler to accept client connections' in Group Policy.
Medium SVC-005 (WARNING)	Security Weakness	**Third-Party Services Running as LocalSystem:** 15 third-party services are running as LocalSystem (including: LenovoVantageService, DAService, EABackgroundService, GameInputRedistService, GamingServices, HidHideWatchdog.exe). Total LocalSystem running services: 91/135.	Audit third-party background services and uninstall unused software or downgrade service accounts to LocalService where supported.

14. Final Security Posture Assessment

Based on the automated read-only inspection of **LAB-DEMO-PC01 (DEMO SAMPLE DATA)** running **Microsoft Windows 11 Home (Build 26200)**, the host achieved an overall security posture score of **86.0 / 100 (Elevated Risk (Action Required))**. Core platform defenses including Windows Defender Real-Time Protection, Tamper Protection, Windows Firewall across all three profiles, UAC Secure Desktop, UEFI Secure Boot, SMBv1 deprecation, and NTFS system directory permissions were verified as active and compliant. Addressing the **3 failed check(s)** and **2 warning(s)** listed in Section 12–13 will bring the workstation into full alignment with security hardening benchmarks.